

12.2 The Reverse Process

MSE on the noise you added. T reverse steps. Lab 12 uses the oracle mean.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Generation walks the forward process **backward**. You start from $x_T \sim \mathcal{N}(0, \mathbf{I})$ and, step by step, remove a bit of noise until you have x_0 .

First time the reverse process appears. The **reverse** is: predict the noise you added, take a step toward x_0 , repeat T times.

What. A net $\varepsilon_\theta(x_t, t)$ predicts ε . Training is MSE on that noise. Sampling starts from $x_T \sim \mathcal{N}(0, I)$ and denoises. **Why.** You cannot write the true reverse in closed form for images. Predicting ε is equivalent (under the usual parameterization) to predicting the posterior mean. **Architecture.** U-Net or DiT on x_t with a time embedding. Lab 12 uses an **oracle** mean so you see the geometry without training. **How.** Loss: $\mathbb{E} \|\varepsilon - \varepsilon_\theta(x_t, t)\|^2$. Sample: T reverse steps (or fewer with a faster sampler, later papers). **Formula.** x_{t-1} from ε_θ via the standard DDPM mean; cartoon: subtract a scaled ε_θ and add a little noise (except $t = 1$). **Tradeoffs.** + MSE is easier than GAN min-max. – Many steps at sample time; quality vs speed; classifier-free guidance is next note.

1. Predict the noise, not the image

Ho et al. (DDPM) train a net $\varepsilon_\theta(x_t, t)$ to guess the ε you added in note **12.1**. The usual loss is

$$\mathbb{E}_{t, x_0, \varepsilon} \|\varepsilon - \varepsilon_\theta(x_t, t)\|^2.$$

Predicting ε is equivalent (up to scaling) to predicting x_0 or the score $\nabla_{x_t} \log q(x_t)$. Pick one parameterization and stay consistent.

Reverse: a network predicts the noise (or x_0) and steps toward data



Training is ordinary regression. Sample t uniformly, form x_t with the one-liner from **12.1**, predict ε , take MSE. No adversary.

2. One reverse step

Given x_t and a noise prediction, a typical mean for x_{t-1} is

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \varepsilon_\theta(x_t, t) \right) + \sigma_t z,$$

with $z \sim \mathcal{N}(0, \mathbf{I})$ for $t > 1$, and σ_t from the schedule (often $\sigma_t^2 = \beta_t$).

If you **knew** x_0 , the true posterior $q(x_{t-1} | x_t, x_0)$ is Gaussian with a closed-form mean. Lab 12 uses that oracle mean for a single step so you can see denoising without training a U-Net.

The oracle is a teaching crutch. A trained ε_θ is what replaces the known ε (and therefore the known x_0) at sample time.

3. Sampling cost

You pay T network calls (or fewer with DDIM / distilled samplers). That is the trade: stable regression training, slow ancestral sampling. GANs (Week 11) sample in one forward pass and fight mode collapse instead.

For a project, report the sampler (T , DDPM vs DDIM) next to the metric. Changing T at test time is part of the method.

4. Teaching this note

About **35 minutes** at the board, then **~6 minutes** of video.

- **0–10 min.** MSE on ε . Same x_t formula as **12.1**.
 - **10–22 min.** Write one reverse-step mean. Circle the ε_θ term.
 - **22–33 min.** Worked scalar reverse step, then the Lab 12 oracle story.
 - **Then** play Umar Jamil **13:00–16:36** (training loop, then sampling loop). Pause when he draws one denoising step.
-

5. Worked example

Reuse $x_0 = 1.0$, $\bar{\alpha}_t = 0.64$, $\varepsilon = 2.0$, so $x_t = 2.0$ as in note **12.1**. Take $\alpha_t = 0.64$ for a single-step cartoon (so $\sqrt{\alpha_t} = 0.8$, $1 - \alpha_t = 0.36$, $\sqrt{1 - \bar{\alpha}_t} = 0.6$).

Suppose $\varepsilon_\theta = \varepsilon = 2.0$ exactly, and drop $\sigma_t z$ for the mean:

$$x_{t-1} = \frac{1}{0.8} \left(2.0 - \frac{0.36}{0.6} \cdot 2.0 \right) = 1.25 (2.0 - 1.2) = 1.25 \times 0.8 = 1.0.$$

The reverse mean lands on x_0 . That is the oracle: if you predict the true noise, one algebraic step peels it off (here $\alpha_t = \bar{\alpha}_t$, a first-step special case). At a generic t you do not jump all the way to x_0 ; you move **closer**. Lab 12 checks $\|x_t - x_0\|$ vs $\|\hat{x} - x_0\|$ at $t = 20$ and wants the second smaller.

If $\varepsilon_\theta = 0$ (the net refuses to denoise), the subtracted term vanishes and you mostly rescale x_t , which is not a sample from the data.

 Oracle reverse mean: $\langle x_t - \varepsilon_\theta \frac{x_t - x_0}{\alpha_t - \bar{\alpha}_t} \rangle$

2.0 peels back to 1.0" src="graphics/12.2-diffusion-reverse/reverse-mean.png" />

Sampling is an **iterative** procedure. You pay T network calls (or fewer with DDIM). A GAN still samples in one pass and fights collapse instead.

6. Where students get stuck

- Training a U-Net in Lab 12. The lab uses the **true** posterior mean because you know x_0 .
- Omitting t as an input to ε_θ . The same x_t is a different problem at $t = 10$ and at $t = 900$.
- Forgetting $\sigma_t z$ on the way down and then wondering why every sample looks identical.

7. Video

Watch [Umar Jamil, How diffusion models work](#), **13:00–16:36**.

Pause on the training objective (predict the noise) and on the sampling loop (many reverse steps). Same URL as note **12.1**; do not replay 7:35–13:00 unless needed.

8. Practice

1. If $\varepsilon_\theta = \varepsilon$ exactly, what should happen to $\|x_{t-1} - x_0\|$ relative to $\|x_t - x_0\|$, in expectation, for mid-range t ?
2. Why add $\sigma_t z$ on the way down instead of always taking the mean?
3. You may not train a score network in Lab 12. What replaces ε_θ for that one reverse step?
4. $x_t = 3$, $\alpha_t = 0.25$, $\bar{\alpha}_t = 0.25$, $\varepsilon_\theta = 2$, no extra noise. Compute the reverse mean. ($\sqrt{\alpha_t} = 0.5$, $\sqrt{1 - \bar{\alpha}_t} = \sqrt{0.75} \approx 0.866$, $1 - \alpha_t = 0.75$.)
5. Training samples t uniformly from $1 \dots T$ and one ε . Why is that enough, instead of a full reverse chain inside the loss?